

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ407			
Course Title	CRYPTOGRAPHY			
Type of Course	Elective			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Elementary number theory, algebra, combinatorics, basic linear algebra			
Course Summary	Cryptography is a fundamental aspect of information security that involves creating secure communication by encoding messages to make them unintelligible to unauthorised users and Cryptography relies heavily on mathematical concepts. This course covers a wide range of topics, starting with Classical Cryptography, which includes simple cryptosystems. It also delves into cryptanalysis of these systems. Moreover, the course includes a section on Cryptographic Hash Functions, focusing on their role in ensuring data integrity. Students gain a comprehensive understanding of these concepts and techniques, equipping them with the knowledge and skills needed to analyze and implement secure cryptographic systems.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Construct the parity check/generator matrix of a linear code. Design cyclic codes of a given rate and distance parameters.	Ap	C	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO2	Calculate bounds on rate and distance of a given linear code using various bounds.	An	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Decode a cyclic code using various standard decoding procedures.	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Cryptography Theory and Practice 3 rd Edition, Douglas R. Stinson, Chapman & Hall			
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I	Classical Cryptography		12	Min.15
	1	Chapter 1: Section 1.1-1.1.1: Some Simple Cryptosystems, Shift Cipher		
	2	Chapter 1: Sections 1.1.2 & 1.1.3: The Substitution Cipher, Affine Cipher		
	3	Chapter 1: Sections 1.1.4 & 1.1.5: The Vigenere Cipher, The Hill Cipher		
	4	Chapter 1: Sections 1.1.6 : The Permutation Cipher		
	5	Chapter 1: Sections 1.1.7 : Stream Ciphers		
II	Cryptanalysis		12	Min.15
	6	Chapter 1: Section 1.2 & 1.2.1 : Cryptanalysis: Cryptanalysis of the Affine Cipher		
	7	Chapter 1: Section 1.2.2 : Cryptanalysis of the Substitution Cipher		
	8	Chapter 1: Section 1.2.3 : Cryptanalysis of the Vigenere Cipher		
	9	Chapter 1: Section 1.2.4 : A known plain text attack on the Hill Cipher		
	10	Chapter 1: Section 1.2.5 : Cryptanalysis of the LFSR-based Stream Cipher.		
III	Shannon's Theory		10	Min.15
	11	Chapter 2 : Sections 2.1, 2.2 : Introduction, Elementary Probability Theory		
	12	Chapter 2 : Sections 2.3: Perfect Secrecy		
	13	Chapter 2 : Sections 2.4: Entropy, Huffman Encodings		
	14	Chapter 2 : Sections 2.5: Properties of Entropy		
	15	Chapter 2 : Sections 2.6: Spurious Keys and Unicity Distance		
	16	Chapter 2 : Sections 2.7: Product Cryptosystems		
IV	Block Ciphers and Advanced Encryption Standard		14	Min.15
	17	Chapter 3: Sections 3.1 and 3.2 : Introduction, Substitution - Permutation Networks		
	18	Chapter 3: Sections 3.3 (3.3.1 to 3.3.3): Linear Cryptanalysis		
	19	Chapter 3: Sections 3.4 : Differential Cryptanalysis		
	20	Chapter 3: Sections 3.5 (3.5.1, 3.5.2) : Data Encryption Standard (DES), Description of DES, Analysis of DES		
V	Open Ended		12	
	Cryptographic Hash Functions			
References	1. Jeffrey Hoffstein: Jill Pipher, Joseph H. Silverman, An Introduction to Mathematical Cryptography, Springer International Edition. 2. Koblitz, N. (1994) A course in Number Theory and Cryptography, (Second Ed.), Springer- Verlag			

	3. Yan, S. Y. (2003) Primality Testing and Integer Factorization in Public-Key Cryptography, Springer 4. H. Deffs & H. Knebl: Introduction to Cryptography, Springer Verlag, 2002 5. Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone: Handbook of Applied Cryptography, CRC Press, 1996. 6. William Stallings: Cryptography and Network Security Principles and Practice, Third Edition, Prentice-hall India, 2003. 7. D. Boneh and V. Shoup: A Graduate Course in Applied Cryptography (V 0.5) 8. J. Katz and Y. Lindell. Introduction to Modern Cryptography (2nd edition)
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Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	3	3	3	0	0	3
CO 2	3	3	1	1	3	3	3	3	0	0	3
CO 3	2	3	2	1	3	3	3	3	0	0	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓